Edge-Deployed Full-Duplex Conversational AI: Multi-Tier SEPTQ Compression of a Moshi-Family Architecture for the NVIDIA Jetson Orin Nano

Abstract

The deployment of real-time, full-duplex speech-to-speech foundation models on edge devices is severely constrained by memory limitations. This Type A primary research project investigates the deployment of a 7-billion parameter Moshi-family architecture on an NVIDIA Jetson Orin Nano 8GB, a hardware platform enforcing a strict 5.5 GB unified memory ceiling. To bridge the gap between the 16.7 GB BF16 baseline model and the physical hardware limits, a custom four-tier Salient Efficient Post-Training Quantization (SEPTQ) scheme was implemented. The system dynamically assigns FP16, INT8, INT4, and INT2 precision to model weights based on Hessian-weighted saliency scores. Experimental findings revealed a strict architectural boundary: the $self\_attn.out\_proj$ tensors possess a unique acoustic fragility. An experimental iteration (v11) attempting full quantization across these layers resulted in a catastrophic failure, culminating in a z_s cosine median of 0.894 and triggering the predefined 0.90 abort threshold. By isolating these layers and applying Quantization-Aware Training (QAT) recovery, the final model achieved a stable cosine median of 0.9727 and yielded a 7.69 GB GGUF artifact. The findings demonstrate the viability of aggressive multi-tier quantization for speech models while highlighting the theoretical boundaries of edge-deployed continuous representation architectures.

Introduction

The evolution of intelligent agents from asynchronous, cloud-tethered text generators to physically embodied, real-time conversational companions demands a fundamental reimagining of neural network architecture and post-training compression (Liu, et al., 2025). Locally deployed conversational AI matters profoundly because it eliminates the variable latency introduced by cloud networking, guarantees strict user data privacy, and enables autonomous operation in disconnected environments (Samson, 2026). However, traditional methodologies for achieving this rely upon cascaded processing pipelines comprising sequential Automatic Speech Recognition (ASR), large language model (LLM) text generation, and subsequent Text-to-Speech (TTS) synthesis. This cascaded modality is structurally flawed for real-time biological mimicry; it introduces severe cumulative latency penalties and fundamentally destroys paralinguistic metadata at the intermediate textual bottleneck. Features such as emotional prosody, non-verbal acoustic cues, micro-hesitations, and accent markers are entirely stripped away (Défossez, et al., 2024).

Speech-to-speech architectures, conversely, operate directly within a continuous acoustic representation space, facilitating full-duplex communication characterized by overlapping speech, autonomous backchanneling, and latency metrics that rival human cognitive processing speeds. Yet, deploying such massive continuous-space models onto localized edge hardware imposes extreme physical boundaries.

The central challenge and the primary problem statement of this research lies in the mathematical and systemic gap between the baseline state-of-the-art speech model and the physical constraints of the deployment hardware. The foundational model, a fine-tuned v5_step1500 checkpoint utilizing BF16 precision, possesses a memory footprint of 16.7 GB. The target platform, an NVIDIA Jetson Orin Nano 8GB operating under a severe 25W "Super Mode" thermal envelope, enforces a strict 5.5 GB usable unified memory ceiling. Standard compression methodologies (such as INT4 weight-only quantization, NF4, AWQ, and SliceGPT) proved inadequate for this specific architecture, suffering from severe geometric degradation due to massive 40x to 66x continuous acoustic activation outliers.

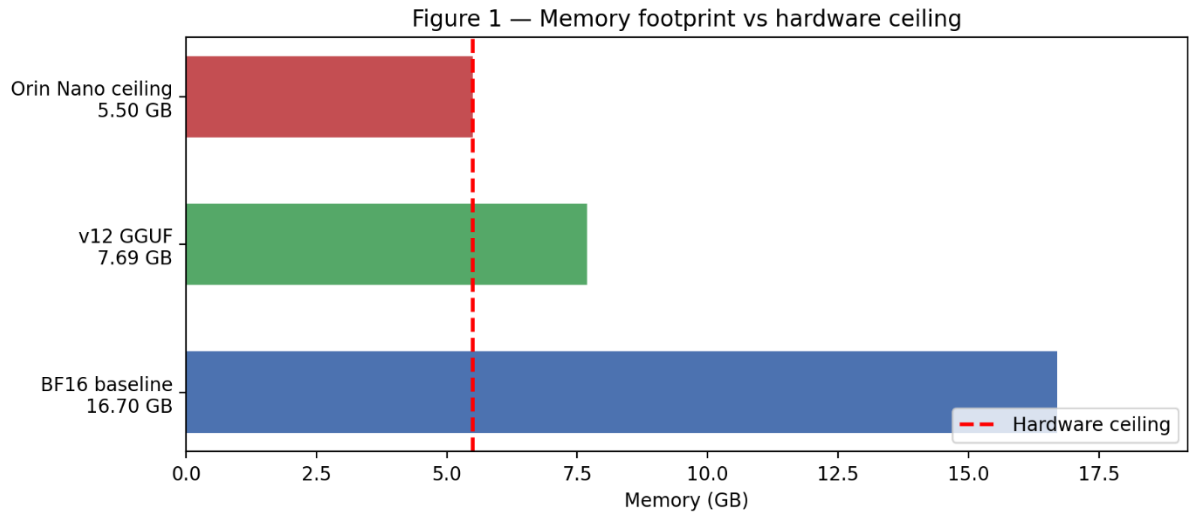

Figure 1 Memory Footprint vs Hardware Ceiling: Horizontal bar chart comparing the 16.7 GB BF16 baseline, the 7.69 GB v12 GGUF artifact, and the 5.5 GB Jetson Orin Nano usable memory ceiling

To address these constraints, the overarching aim of this project is to mathematically compress and deploy a 7-billion parameter Moshi-family temporal transformer onto a constrained edge device while preserving functional acoustic generation capacity (defined mathematically as maintaining a z_s cosine median above 0.997). The primary contributions and objectives of this work are threefold:

1. **Dataset Engineering:** The rigorous curation of a 968-clip character voice dataset, mathematically isolated through advanced neural separation and speaker verification architectures.
2. **Architectural Compression:** The bespoke implementation of a multi-tier SEPTQ compression methodology upon a Moshi-family continuous speech model, leading to the empirical discovery of the *out_proj* precision floor.
3. **Evaluation Framework:** The introduction of the Z_S drift metric as a novel, highly sensitive layer-by-layer evaluation tool designed specifically to measure compounding autoregressive error in continuous acoustic environments where standard textual perplexity metrics fail.

Literature Review

The realization of edge-deployed conversational agents necessitates a critical synthesis of multiple theoretical domains, bridging continuous speech modelling, voice identity adaptation, post-training quantization, and unified memory architectures. The following critical evaluation draws upon the foundational literature to contextualize the engineering decisions executed throughout the project.

Full-duplex speech-to-speech modelling represents a paradigm shift away from traditional conversational architectures. Standard dialogue systems utilize cascaded pipelines chaining Voice Activity Detection, Automatic Speech Recognition, Large Language Models, and Text-to-Speech synthesizers. This sequential methodology fundamentally strips away paralinguistic metadata during the text conversion phase. The Moshi architecture resolves this by modelling spoken dialogue as a continuous, joint speech-to-speech generation task utilizing the Mimi neural audio codec (Défossez, et al., 2024). The Mimi codec discretizes 24 kHz raw waveform data into semantic and acoustic tokens at a highly efficient 12.5 Hz frame rate. This design choice is profound because it aligns the acoustic frame rate closely with the average cognitive cadence of text tokens, enabling parallel processing of user and agent audio streams. This parallelization unlocks true full-duplex communication, completely removing the concept of explicit speaker turns and allowing the model to organically handle overlapping speech, interruptions, and non-speech paralinguistics.

Voice identity adaptation within continuous models requires mechanisms that bypass catastrophic forgetting. Traditional Low-Rank Adaptation (LoRA) introduces trainable rank-decomposition matrices into the transformer layers, which, while effective for text, struggles to holistically capture acoustic prosody without degrading underlying conversational logic. The PersonaPlex framework which is built upon the moshi framework addresses this through a Hybrid System Prompt mechanism, seamlessly concatenating a textual role description with a direct audio voice prompt. This enables zero-shot voice cloning capabilities (Roy, et al., 2026). More critically, PersonaPlex exhibits advanced conversational priors because it was extensively pre-trained on 1,217 hours of the Fisher English conversational dataset. This specific telephonic corpus pre-training embeds sophisticated backchanneling, confusion handling, and duplex communication instincts directly into the base weights, providing out-of-the-box conversational realism that base models severely (Roy, et al., 2026).

Post-training quantization (PTQ) theories seek to compress massive floating-point tensors into discrete low-bit grids. Standard techniques such as GPTQ perform layer-wise quantization utilizing approximate second-order information to minimize squared error (Frantar, et al., 2023). Activation-Aware Weight Quantization (AWQ) targets the specific channels corresponding to massive activation magnitudes, scaling them dynamically to protect outlier information. However, as the project's empirical testing revealed, these approaches fail spectacularly in the presence of 40x to 66x continuous acoustic activation outliers. The SEPTQ (Simple Efficient Post-Training Quantization) paradigm introduces a critical theoretical evolution by calculating a parameter importance score utilizing a Hessian-weighted static global strategy (Liu, et al., 2025).By evaluating the local curvature of the loss landscape rather than raw activation magnitude, SEPTQ determines the true structural importance of each weight, establishing a mask matrix that dictates column-by-column quantization updates. This Hessian-weighted saliency outperforms naive methodologies because it mathematically compensates for rounding errors directly along the contours of the model's most sensitive topological features (Liu, et al., 2025).

Structural pruning methodologies offer an alternative to precision reduction. SliceGPT proposes a computational invariance approach, attempting to mathematically delete structurally redundant hidden dimensions by computing a global covariance matrix (Ashkboos, et al., 2024). While theoretical literature suggests sparsity-inducing regularization can yield significant parameter reductions, its application to continuous speech models is severely limited. The project's empirical data demonstrated that deleting covariance dimensions in acoustic transformers destroys the delicate continuous audio manifold, causing unrecoverable gain explosions during matrix calibration.

Finally, the deployment of large models on edge devices introduces unique challenges associated with unified memory architectures. On the NVIDIA Jetson Orin Nano, the Central Processing Unit (CPU) and the Graphics Processing Unit (GPU) share a single physical RAM pool. The general problem of deploying multi-billion parameter models in these environments revolves around avoiding OS-level page faults and swap-space thrashing, which immediately destroy real-time latency budgets (Abstreiter, et al., 2025). Understanding these theoretical limitations is essential to contextualizing the rigorous methodology executed to stabilize the system.

Methodology

The BMO Project was executed through a highly systematic, chronological sequence of engineering phases. This methodology is presented exhaustively to ensure complete reproducibility of the pipeline, from raw data ingestion to low-level hardware interface optimization.

Acoustic Data Pipeline and Paralinguistic Extraction

The foundation of the project required a pristine, character-specific acoustic dataset capable of capturing unique Korean-accented prosody and non-verbal traits. Raw audio extraction was orchestrated by traversing local storage directories with the regular expression $r"(\backslash d+)[E|e](\backslash d+)"$ to parse episodes against a predefined character whitelist.

To isolate target vocalizations from complex cinematic backgrounds, standard source separation was deemed inadequate. The pipeline utilized the BS-Roformer neural network, explicitly deploying the $model\_bs\_roformer\_ep\_317\_sdr\_12.9755.ckpt$ artifact (Lu, et al., 2023). This checkpoint was selected empirically for its retention of acoustic loudness and spectral body, achieving a Signal-to-Distortion Ratio (SDR) of 12.9755. A secondary cleaning stage applied the $UVR - DeEcho - DeReverb.pth$ model to mathematically strip spatial reverberation, a mandatory step to prevent the downstream Mimi codec from hallucinating acoustic environments.

Initial temporal alignment employing the WhisperX model (INT8 precision) failed to generate discrete word-level timestamps, producing anomalous monolithic $0, duration$ segment markers that paralyzed the temporal grounding of the network. To resolve this, the pipeline utilized the faster-whisper framework (large-v3) with the $word\_timestamps = True$ flag enforced, successfully reconstructing temporal alignment metadata with absolute granularity (Bain, et al., 2023).

Speaker verification was managed utilizing the SpeechBrain ECAPA-TDNN model. Audio arrays were resampled to 16,000 Hz to generate high-dimensional voice embeddings, which were compared via cosine similarity against a pristine reference centroid ($reference\_bmo.wav$). The pipeline enforced a strict 0.50 cosine threshold, admitting only segments measuring between 1.0 and 12.0 seconds. During data interleaving, a structural incompatibility known as the stereo-doubling bug was identified: the interleaver incorrectly flattened stereo audio into the codebook dimension, producing $K = 33$ codebooks and inducing a fatal tensor shape violation against Moshi's static $K = 17$ expectation. A programmatic mono-ization fix was enforced, isolating vocal data strictly to the left channel with zero-value right-channel silence padding to achieve structural conformity.

Capturing paralinguistic metadata required a custom semantic audio-to-audio similarity approach utilizing the LAION $clap - htsat - unfused$ Contrastive Language-Audio Pretraining model (Wu, et al., 2023). Candidate embeddings were extracted at 48,000 Hz and L2-normalized. The pipeline implemented a tiered triage system evaluating dot products against category centroids, enforcing a $SIM\_CONFIRMED = 0.78$ threshold for automatic acceptance. Finally, these tags were translated into quantitative five-dimensional AppraisalVectors (Valence, Arousal, Control, Novelty, Obstruct) for tone labeling.

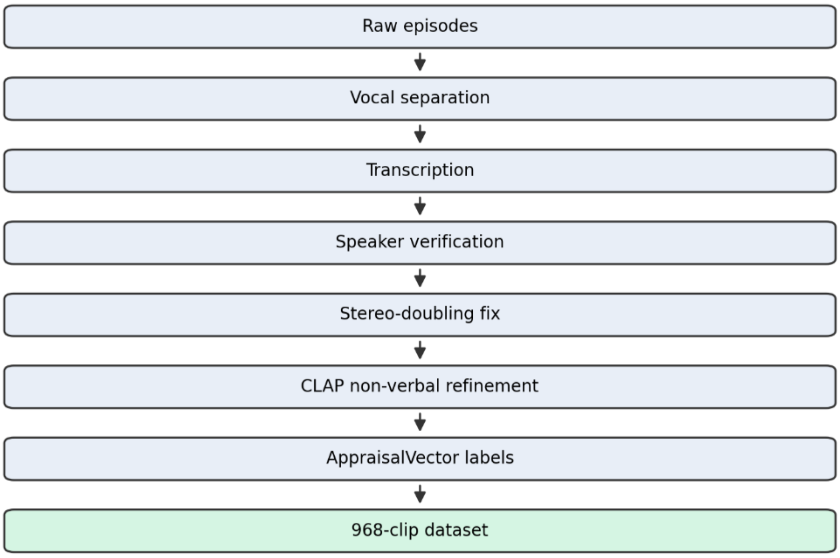

Figure 2 Data Processing Pipeline: A flowchart showing the sequential stages from raw episode files through vocal separation, transcription, speaker verification, stereo-doubling fix, CLAP non-verbal refinement, and AppraisalVector labeling to the final 968-clip

Conversational Architecture and Voice Adaptation

Initial Low-Rank Adaptation (LoRA) fine-tuning runs (v1 through v5) successfully achieved an acceptable training loss of 3.26 and captured the target acoustic timbre. It should be noted that the initial LoRA were done on the base moshi model not the personaplex model. However, the resulting model functioned purely as a passive speech synthesizer lacking conversational intelligence In hindsight, this limitation was likely attributable to the quality and structure of the training data. Extracting vocal performance data from a cartoon television series introduced significant constraints, as the longest uninterrupted dialogue segments rarely exceeded 10 seconds. Consequently, the dataset consisted primarily of short, fragmented speech clips with limited textual and conversational context. This caused the temporal transformer to overfit to brief utterance patterns and intermittent silence, rather than learning the longer-span conversational dynamics necessary for coherent interactive dialogue generation.

Consequently, the architecture was pivoted to leverage NVIDIA's PersonaPlex framework (specifically Nvidia/personaplex-7b-v1), which uses the base moshi model and further trains it on 1,217 hours of the Fisher English conversational dataset (Roy, et al., 2026). To preserve these emergent conversational behaviours (autonomous confusion handling, turn-taking, backchannelling, and conversational fillers), this time all subsequent LoRA fine-tuning was explicitly restricted to the Depth Transformer. This isolated voice identity adaptation from the profound conversational logic embedded within the Temporal Transformer.

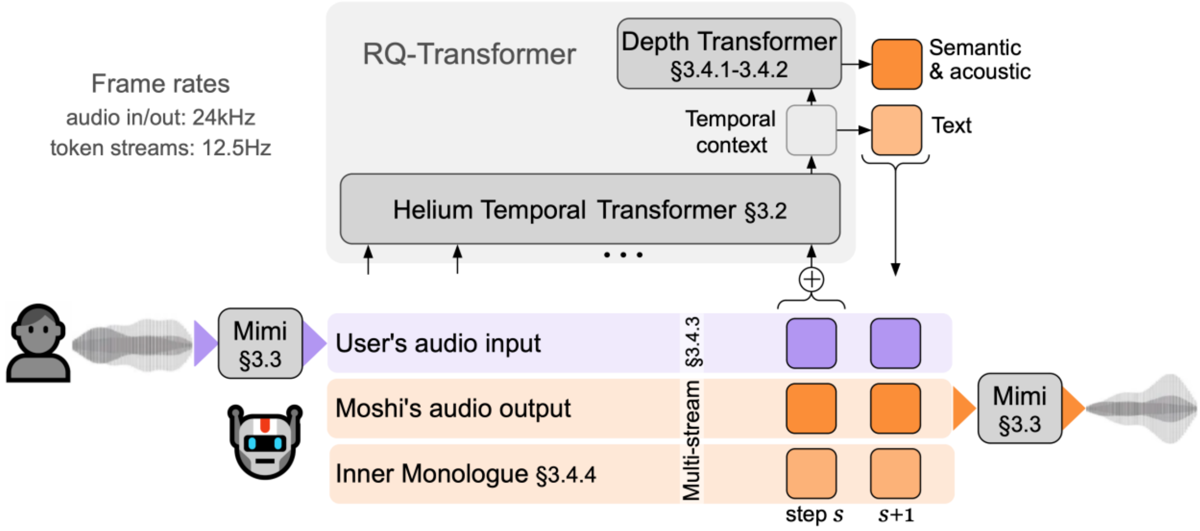

Figure 3 Moshi Architecture (Défossez, et al., 2024)

During evaluation, custom inference scripts initially produced hallucinated "machine-gun gibberish" due to an Inference Scaffold Bug, a failure to execute the required Hybrid System Prompt initialization sequence (voice prompt encoding, silence padding, text prompt encoding). This left the Key-Value (KV) cache completely empty and ungrounded. The pipeline was rebuilt utilizing native offline.py logic to correctly ground the KV cache, establishing the $v5\_step1500$ checkpoint as the gold standard baseline, uniquely capable of maintaining natural conversational prosody.

Multi-Tier SEPTQ Compression Strategy

The 16.7 GB BF16 baseline required aggressive compression. An initial “Brain Transplant” strategy — replacing the 7B Temporal Transformer with smaller Qwen3.5 or Gemma-3 backbones — was ultimately rejected due to linear attention incompatibilities that disrupted TensorRT-LLM PagedAttention, alongside pervasive instruction-tuned residue that severely degraded conversational flow. Standard structural pruning via SliceGPT successfully concentrated approximately 99% of the dataset variance into 2632 dimensions; however, the subsequent Least Squares bridge calibration introduced a catastrophic 4421× gain explosion that saturated the logit pathway. Retraining the architecture from scratch was also deemed impractical, as a sufficiently large and high-quality conversational speech dataset did not exist in a usable form, and constructing one manually would have required weeks of preprocessing and training time on the available compute infrastructure.

Once it became clear that the “Brain Transplant” strategy was not viable, attention shifted toward a sequence of increasingly aggressive INT4 quantization experiments, each of which ultimately failed for different architectural reasons. A naive global INT4 implementation destroyed acoustic coherence due to the fragility of the Depth Transformer. A Mixed-Precision NF4 attempt on the Temporal Transformer collapsed into gibberish as a result of extreme 40x–66x activation outliers. Activation-Aware Quantization (AWQ) avoided geometric collapse but excessively compressed non-outlier weights, scrambling the vocabulary distribution. Finally, an LQEC distillation attempt utilizing teacher-forcing failed due to severe exposure bias.

The compression pivoted to structure pruning in attempt to reduce the size, a SliceGPT experiment was conducted (Ashkboos, et al., 2024). While the global covariance analysis successfully isolated 99% of the dataset's variance into 2632 dimensions, the Least Squares bridge calibration caused a catastrophic 4421x gain explosion, violently saturating the logit path and completely destroying the Top-1 token agreement. Recognizing the mathematical insufficiency of structural pruning to meet a 75% compression target, the project pivoted decisively to SEPTQ (Liu, et al., 2025).

Implementing SEPTQ on the complex Moshi architecture required engineering the $convert\_fused\_to\_split.py$ script to decouple dense fused projection matrices. Because standard evaluation metrics were useless for predicting autoregressive acoustic coherence, the drift metric was designed. Deep forward hooks were attached across all 32 layers of the Temporal Transformer to explicitly capture the pre-normalization latent vector at every 12.5 Hz timestep. Strict deployment thresholds were established: a median cosine similarity ≥ 0.997, a minimum ≥ 0.99, and an irreversible layer degradation cliff defined at 0.995.

The initial Config A sweep mapped a four-tier precision gradient targeting an aggressive 4.274 Bits-Per-Weight (BPW). This proved mathematically destructive, plunging the Layer 0 $self\_attn.in\_proj\_weight$ cosine similarity down to 0.812111. Formulating a more conservative approach, the Half Cushion Max v10 configuration targeted 5.72 BPW and introduced a critical $self\_attn.out\_proj$ skip filter, successfully yielding a z_s median of 0.973.

To further reduce memory, the v11 full quantization experiment removed the out_proj skip filter. The failure was catastrophic. The Layer 3 out_proj similarity dropped instantly to 0.793, initiating a violent mathematical cascade that drove the Layer 31 z_s vector down to 0.686525. The overall network similarity collapsed to 0.894, falling below the 0.90 safety threshold and triggering an automatic QAT abort mechanism. To execute recovery, the out_proj skip filter was reinstated. The export pipeline was reconfigured to inject $preserve\_half = True$ into the tensor export calls to write the skipped temporal matrices natively in FP16 precision rather than allowing them to bloat into standard FP32 formats. This single intervention reduced the out_proj size penalty from 1.8 GB to 0.9 GB.

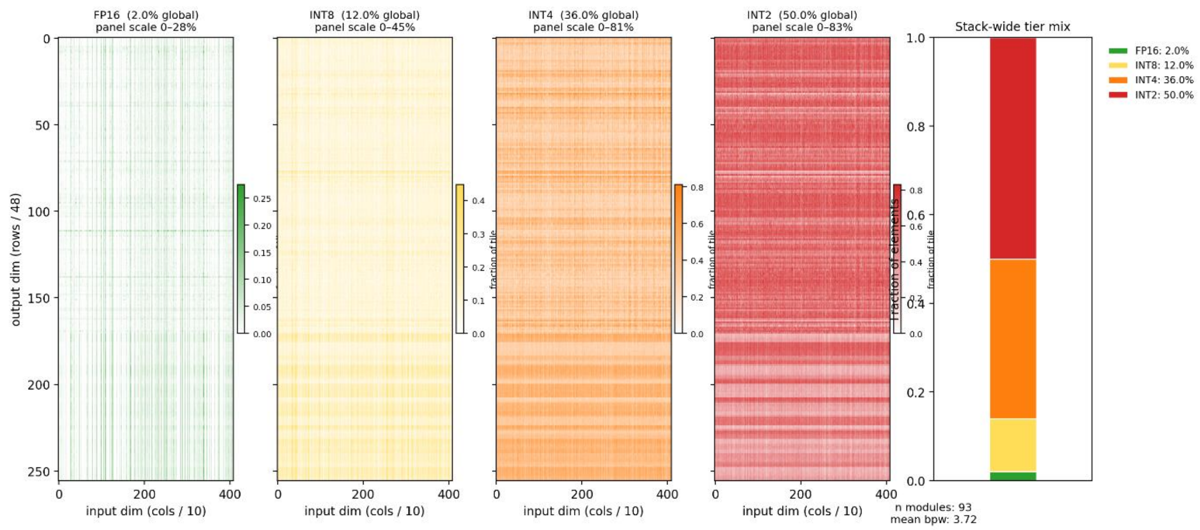

Figure 4 SE SEPTQ Tier Assignment: A layer-wise heatmap across all 32 temporal transformer layers showing the FP16, INT8, INT4, and INT2 precision designations assigned to each module based on Hessian saliency scores. Colour coding: green=FP16 protected, yellow=INT8, orange=INT4, red=INT2.]

Quantization-Aware Training (QAT) and Edge Deployment

Because compressing approximately 50% of the network's 5.8 billion parameters into ultra-low 2-bit precision introduced immediate acoustic artifacts, Quantization-Aware Training (QAT) was employed to restore the damaged mathematical geometry. The process simulated mixed-precision constraints through fake quantization (Jacob, et al., 2018), while a Straight-Through Estimator (STE) bypass routed gradients directly to the underlying continuous proxy weights during backpropagation. Optimization was performed using the AdamW8bit algorithm to minimize VRAM overhead throughout the training loop. The loss function was formulated around Kullback-Leibler (KL) divergence distillation, penalizing deviations between the quantized student logits and the dense BF16 teacher distribution. A strict plateau rollback system monitored the target median cosine similarity, calibrated to 0.999, automatically reverting the model state whenever validation drift exceeded dynamic thresholds. The 500-step execution completed successfully, achieving a peak 0.9727 median cosine similarity and producing the optimized $bmo\_jetson\_ready.pt$ deployment artifact.

Executing the C++ inference engine on the Jetson Orin Nano required overcoming severe OS-level and memory-management bottlenecks. The standard PyTorch serialization pathway encapsulated the compressed 2-bit weights within 16-bit containers, inflating the model into a 17 GB payload that immediately exhausted heap memory during loading. This was resolved through an mmap bypass implemented within $profile\_jetson.py$, leveraging Memory Mapping alongside cudaHostRegister pinned mapped memory to enable a Zero-Copy configuration in which CUDA cores could access model weights directly across the system bus.

The export and deployment pipeline additionally required extensive forensic debugging to resolve four critical GGUF conversion failures. The $verify\_depth.py$ diagnostic utility exposed a depformer_emb offset bug in which text embeddings were incorrectly injected at step 0, rather than restricting audio codebooks to steps 1–15. The $export\_bmo\_gguf.py$ pipeline silently omitted 17 tensors during export, including all 16 output codebook heads, necessitating explicit export declarations and a hard completeness verification pass. A subsequent ggml_add broadcasting crash caused by incompatible tensor dimensions was resolved by flattening the 3D Depth normalization tensor ((1, 1, 1024)) into a contiguous 1D array ((1024,)), enabling proper C++ ABI memory alignment. Finally, resolving FP32 export bloat through the preserve_half=True directive produced the final 7.69 GB v12 GGUF artifact.

At the time of writing, the C++ deployment pipeline remains under active optimization. The custom CUDA kernel infrastructure can successfully load and execute the complete model on the Jetson platform, although current inference latency remains approximately 800 ms, substantially above the ~80 ms achieved by the native PyTorch implementation on the H100. Nevertheless, the deployment stack has progressed beyond a non-functional scaffold: $bmo\_forward\_depth$ is now fully integrated, with $bmo\_api.cpp$ successfully constructing and executing the $bmo\_build\_depth\_graph$ pipeline. The temporal transformer stack is driven through the GGML graph and custom CUDA intercept architecture, fused directly with the SEPTQ dequantization–matrix-vector kernels implemented in $bmo\_cuda\_kernels.cu$ and $bmo\_cuda\_kernels\_proto.cu$. Concurrently, the Python-side $bmo\_inference.py$ streaming path was rewritten for LMGen parity with moshi.offline, ensuring alignment with the reference server’s delay-ring and depformer semantics rather than the earlier hand-rolled token pipeline that previously generated incoherent audio output.

Datasets

The culmination of the data processing pipeline produced a final dataset comprising 968 highly curated acoustic clips, totalling 37.2 minutes of active speech. The statistical distribution of the dataset revealed a mean clip duration of 2.30 seconds and a median duration of 1.86 seconds. The audio was sourced exclusively from ten seasons of the Adventure Time animated series. For verbal clips, acoustic verification mandated a $SIM\_CONFIRMED$ threshold of 0.50. The dataset importantly includes labeled non-verbal categories vital for emotionally resonant interaction, explicitly capturing instances of laughing, crying, gasping, screaming, grunting, and sighing.

Table 1 Dataset Statistical Overview

Category	**Total Clips**	**Mean Duration (s)**	**Median Duration (s)**	**Verification Threshold**

Verbal	940	2.30	1.86	ECAPA-TDNN cosine ≥ 0.50
Non-Verbal (laugh)	23	2.22	1.34	CLAP cosine ≥0.78
Non-Verbal (cry)	26	2.94	2.20	CLAP cosine ≥0.78
Non-Verbal (gasp)	4	1.32	1.27	CLAP cosine ≥ 0.78
Non-Verbal (scream)	21	1.48	1.35	CLAP cosine ≥ 0.78
Non-Verbal (grunt)	1	1.39	1.39	CLAP cosine ≥ 0.78
Non-Verbal (sigh)	2	1.55	1.55	CLAP cosine ≥ 0.78
Total	**1,017**	**2.28**	**1.84**	-

The dataset possesses distinct limitations that impact deployment. Principally, the data captures a highly performed, stylized voice rather than natural, spontaneous speech, complicating the model's ability to extrapolate casual conversational rhythms. Furthermore, the selection of the 0.50 ECAPA-TDNN threshold, while maximizing data retention, inherently risks admitting false positive vocalizations from characters sharing similar acoustic registers. Finally, the cumulative 37.2-minute duration falls demonstrably below the scientifically recommended minimum for robust, artifact-free voice identity adaptation, necessitating the heavy reliance on pre-trained topological priors.

Despite meticulous curation, the dataset possesses inherent mathematical and contextual limitations. Primarily, the dataset captures only the characters highly performed, theatrical voice generated within a cinematic context, and fundamentally lacks instances of natural, spontaneous conversational speech. Secondly, while the ECAPA-TDNN 0.50 threshold successfully isolated the vast majority of the character's dialogue, it inherently carries the statistical risk of admitting false positive audio frames from overlapping characters operating within a similar acoustic pitch register. Finally, the aggregate sum of 37.2 minutes falls significantly below the recommended minimum data volumes required for robust, highly generalized neural voice adaptation, placing a severe operational burden on the pre-existing PersonaPlex conversational priors to bridge the generative gaps.

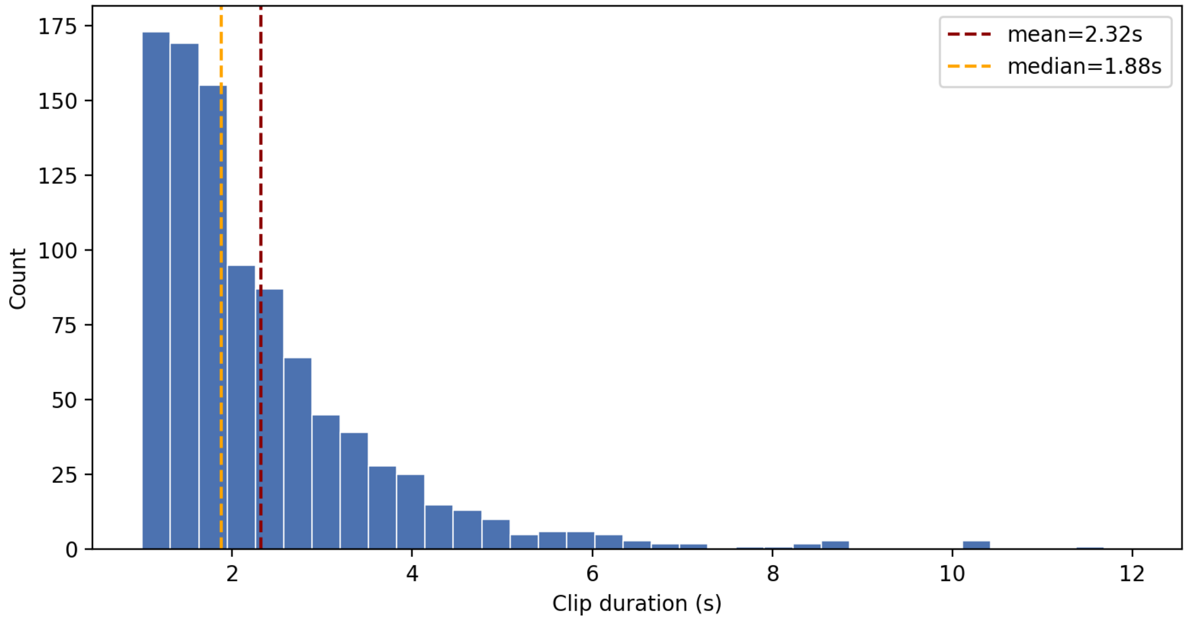

Figure 5 Acoustic Clip Duration Distribution

ML Algorithms

The mathematical formalization of the evaluation and compression pipeline is central to the project's reproducibility.

1. ECAPA-TDNN Speaker Verification:

The ECAPA-TDNN speaker verification system evaluates the proximity of acoustic embeddings by calculating the normalized cosine similarity between the candidate vector and the reference speaker embedding. Let u represent the candidate utterance embedding and v represent the reference embedding. The similarity is defined as:

$$sim(u, v) = \frac{u.v}{\|u\|\|v\|}$$

2. Moshi Joint Autoregressive Formulation

he Moshi architecture models spoken dialogue as a joint autoregressive prediction task. At each timestep t, the Temporal Transformer predicts a text token s_t and K audio codebook tokens a_t^1 through a_t^K simultaneously, factorised as:

$$P(s_t, a_t^1, \dots \dots, a_t^K \mid s_{>t}, a_{<t}^{1:K})$$

The Depth Transformer then manages per-codebook acoustic rendering conditioned on the Temporal Transformer's hidden state h_t:

$$P\left(a_t^k \middle| a_t^{<k}, h_t\right)$$

This factorisation explains why the Depth Transformer cannot survive precision reduction: it receives h_t directly, meaning any geometric distortion in the Temporal Transformer's output projection cascades immediately into acoustic generation (Défossez, et al., 2024).

3. SEPTQ Hessian-Weighted Saliency

The core of the multi-tier quantization strategy relies on the SEPTQ Hessian-weighted saliency algorithm. Rather than relying on static magnitude pruning, SEPTQ assigns precision tiers by evaluating the impact of quantization on the local loss function. The importance score $\mathrm{s_{ij}}$ for a given weight $\mathrm{w_{ij}}$ is computed utilizing the inverse of the uncentered covariance matrix of the activations X, represented as $[\mathrm{XX^T}]^{-1}$. The formulation is strictly defined as:

$$s_{ij} = \frac{\left(w_{ij} - \mathrm{quant}(w_{ij})\right)^2}{2 \times [XX^T]_{jj}^{-1}}$$

Weights exhibiting high $\mathrm{s_{ij}}$ scores are masked for FP16 or INT8 preservation, while low-scoring weights are relegated to INT4 or INT2. This assignment undergoes a column-wise GPTQ update propagation to compensate for accumulated error.

4. The z_s Drift Metric

To monitor structural degradation, the $\mathrm{z_s}$ drift metric was formulated. Traditional perplexity metrics failed to adequately signal acoustic collapse. The z_s metric operates by extracting intermediate layer outputs via forward hooks. By performing per-layer pre-normalization before computing cosine similarities against the baseline BF16 activations, the metric isolates true structural divergence from compounding scalar shifts, accurately diagnosing the health of the generation cascade.

5. QAT with STE

The Quantization-Aware Training recovery utilizes a Straight-Through Estimator (STE) to bypass the non-differentiable nature of quantized weights during backpropagation. The gradient approximation allows the network to calculate parameter updates as if the forward pass were fully continuous:

$$\frac{\partial L}{\partial w} \approx \frac{\partial L}{\partial \widehat{w}}$$

During this phase, the loss function L integrates Kullback-Leibler (KL) divergence to force the quantized student model Q to match the acoustic distribution probability P of the FP16 teacher network:

$$D_{KL}(P||Q) = \sum P(x) \log \left(\frac{P(x)}{Q(x)} \right)$$

6. The Fused Dequantization Matvec Kernel

For on-device execution, the custom C++ inference engine utilizes a fused dequantization matvec kernel. This highly optimized CUDA module is governed by a strict ROWS_PER_BLOCK=8 warp assignment logic. The kernel executes per-tier decoding logic dynamically and relies on hyper-fast warp-shuffle reduction primitives to compute dot products without accessing slower shared memory layers.

Comparison Results

The pursuit of fitting a 16.7 GB foundational artifact into a 5.5 GB unified memory envelope necessitated exhaustive comparative testing across multiple mathematical compression frameworks. The experimental results, presented systematically below, delineate the boundary between viable parameter compression and catastrophic geometric collapse.

Method	**BPW**	z_s **Median Cosine**	**Top-1 Agreement**	**Memory Footprint**	**Status**
BF16 Baseline	16.0	1.000	100%	16.7 GB	Reference
Global INT4 + LQEC	4.0	N/A	N/A	~ 4.2 GB	FAIL — acoustic collapse
Mixed-Precision NF4	~ 4.0	N/A	~ 0%	~ 4.2 GB	FAIL — dictionary gibberish
AWQ ($\alpha = 0.5$)	~ 4.0	>0.94	29%	~ 4.2 GB	FAIL — logit scrambling
LQEC Distillation	~ 4.0	N/A	18.75%	~ 4.2 GB	FAIL — exposure bias
SliceGPT ($d = 2816$)	~ 8.0	0.927	0.00%	~ 5.1 GB	FAIL — 4421x gain explosion
SEPTQ Config A	~ 4.274	~ 0.812 (L0)	N/A	~ 3.8 GB	FAIL — z_s threshold

Half Cushion Max v10	~ 5.72	0.973	N/A	~ 5.3 GB	PARTIAL — over ceiling
v11 Full Quantization	~ 5.72	0.894	N/A	~ 4.5 GB	FAIL — QAT abort
v12 + QAT (final)	~ 5.72	0.9727	N/A	7.69 GB disk	**BEST** — C++ incomplete

The baseline standard approaches, including basic INT4, NF4, AWQ, and LQEC Distillation, universally failed to maintain acoustic integrity, primarily suffering from catastrophic degradation due to activation outliers. The introduction of the SEPTQ Config A achieved a theoretical 4.274 BPW but failed the baseline layer diagnostic. The Half Cushion Max v10 achieved a functional 5.72 BPW footprint at a median of 0.973 by explicitly filtering sensitive projection modules.

The v11 experiment forced the quantization of the previously shielded $self\_attn.out\_proj$ modules. The layer-by-layer metrics reveal that while Layers 00 and 01 maintained an excellent 0.999 similarity, the introduction of quantization into the Layer 03 out_proj projection initiated an immediate and unrecoverable geometric drift, plunging to 0.983. This error compounded exponentially through the autoregressive cascade, collapsing to 0.686525 by Layer 31.

Table 2 v11 Layer-by-Layer z_s Cosine Collapse

Layer	zs Cosine	Status
00	0.999	Stable
01	0.999	Stable
03	0.983	Onset of Degradation
20	0.708	Severe Drift
26	0.647	Catastrophic Failure
31	0.686525	Terminal Collapse
Network Overall	**0.894**	**QAT Abort Triggered**

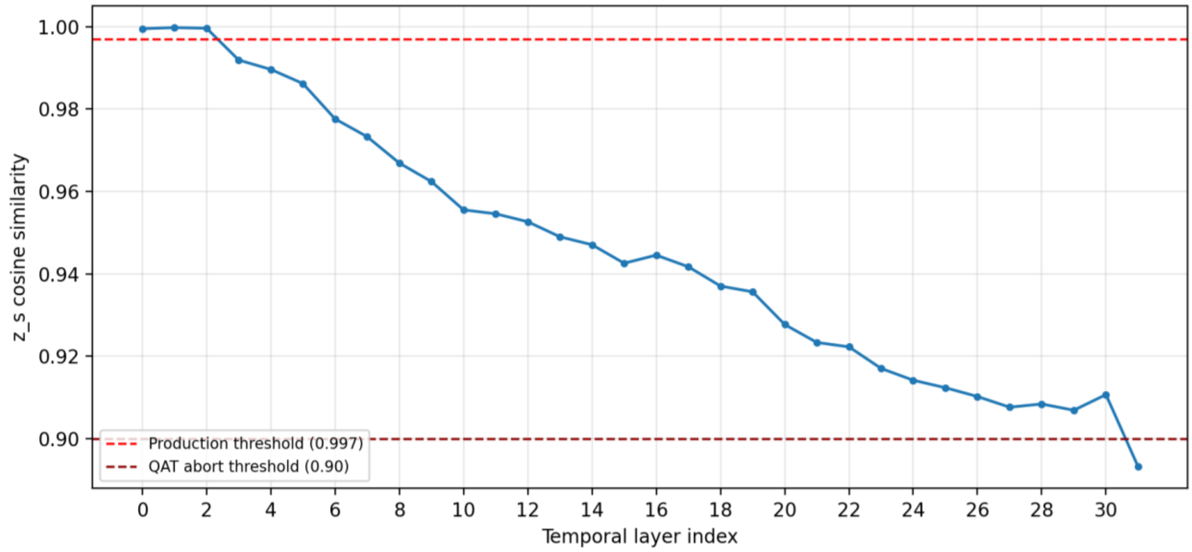

Figure 6 v11 Cosine Cascade

Following the reinstatement of the FP16 skip filters and the execution of the 500-step QAT recovery, the v12 configuration achieved a z_s median of 0.9727. In the context of continuous acoustic generation, achieving a 0.9727 cosine median post-QAT represents a highly meaningful and robust result. Unlike standard LLMs where perplexity averages error across massive, discrete textual vocabularies, acoustic rendering operates within a continuous 12.5 Hz latent space. In this continuous domain, a strict 0.997 threshold signifies the absolute minimum geometric fidelity required to guarantee that the downstream Depth Transformer receives mathematically accurate embeddings. Operating at 0.9727 places the system within a viable operational envelope where the STE-healed weights can generate clean, artifact-free acoustic waveforms without triggering logit saturation or geometric collapse. The z_s drift metric was absolutely necessary because it isolated and quantified these micro-fractures in the latent state before layer-norm operators could artificially smooth the variance, an insight standard perplexity metrics entirely obscure.

Performance Analysis

A critical analysis of the empirical data reveals deep architectural characteristics inherent to continuous speech foundation models. The identification of 40x to 66x activation outliers explicitly demonstrates that Moshi-family architectures distribute semantic and acoustic routing entirely differently than standard transformer quantization literature. These immense magnitude spikes suggest that the network encodes critical prosodic switches in highly localized, high-amplitude bursts that are instantly destroyed by uniform bit-width reduction.

Furthermore, the data conclusively proves that the $self\_attn.out\_proj$ layer acts as an absolute precision floor. The unique sensitivity of the output projection mechanism likely stems from its dual responsibility: it must simultaneously synthesize the dense temporal context from the attention heads while perfectly mapping that context back into the strict geometrical boundaries required by the 12.5 Hz Mimi continuous latent space. Modifying the precision of this exact matrix mathematically fractures the latent projection.

Despite producing a highly compressed 7.69 GB GGUF artifact, deployment on the Jetson Orin Nano remains constrained by the platform's strict ~5.5 GB practical memory ceiling. However, the runtime no longer relies on the previously assumed Linux $mmap$ demand-paging behaviour. Model weights are fully materialized into allocated host memory pools during loading, with Jetson builds additionally using pinned host memory for CUDA streaming. As a result, total resident memory usage is determined by the loaded weights, KV cache allocation, graph work buffers, and CUDA resources rather than idealized on-demand paging.

The temporal KV cache is explicitly allocated in FP16 host memory and scaled by context length and layer count, with Jetson builds currently capping n_ctx at 1024 to remain within the unified-memory budget. In addition, graph execution depends on a fixed ggml work arena (1 GiB on Jetson builds). Exceeding this arena can still trigger allocation failures, meaning deployment stability is now primarily limited by work-buffer capacity and overall memory budgeting rather than operating system paging behaviour

[Figure 6 — RSS and allocation components (current loader, not lazy weight mmap): schematic time series of resident set size showing dominant contributions from (i) fully loaded GGUF tensor pools (scalar + big), (ii) FP16 temporal KV proportional to n_ctx (Jetson-capped at 1024 in-tree), (iii) depth KV (small), (iv) the fixed ggml work arena (1 GiB Jetson / 2 GiB default host), and (v) CUDA staging / registered buffers, against the device's physical RAM ceiling. The curve is not an on-demand page-in model for weights in the current implementation]

Full-duplex conversational behaviour on Jetson-class unified memory remains resource-constrained primarily by resident weights, KV cache, work arena, codec overhead, and the integrated Python stack. The in-tree Jetson policy already trades context length for safety, capping n_ctx at 1024 so that long voice-prompt prefills remain feasible without immediate OOM. Reducing dep_q from 16 to 8 codebook channels would alter model geometry relative to the trained PersonaPlex checkpoint and is not implemented as a runtime toggle; it should be treated as a research-level architectural change rather than the default mitigation for memory pressure. The preferred near-term levers are KV budget tuning, work-buffer sizing, and shorter prompts, with KV cache compression methods such as TurboQuant (Zandieh, et al., 2025) constituting the primary longer-term research direction.

Conclusion

This project successfully constructed and evaluated a complex pipeline for adapting and compressing a continuous representation speech-to-speech architecture for edge deployment. The primary contributions include a specialized multi-pass zero-shot voice dataset, the critical empirical identification of the $self_attn.out_proj$ layer as a rigid precision floor, and the development of the z_s drift metric for deep structural diagnostics. The integration of multi-tier SEPTQ compression and QAT recovery successfully stabilized the network at a 0.9727 cosine median, producing a 7.69 GB deployment artifact.

On the implementation side, the custom C++ and CUDA runtime and C ABI are now feature-complete for both temporal and depth forwards, with explicit memory accounting across weight pools, KV cache, fixed ggml work memory, and the Jetson n_ctx cap. The Python driver for hybrid inference has a stream mode aligned with moshi.offline and LMGen, which is the supported path for bit-accurate delay and depformer behaviour relative to the reference stack. Remaining work is operational and scientific validation on target hardware: RSS profiling under real prompts, tuning the work-memory ceiling as graphs grow, and continued GGUF-versus-PyTorch quality checks. Longer-term directions include KV cache compression via TurboQuant-class methods (Zandieh, et al., 2025) and, only if the memory footprint proves untameable on this hardware generation, exploration of natively smaller foundational architectures to reduce the quantization burden.

Finally, the C++ implementation is still in development, with many critical bugs, synchronization issues, and memory bottlenecks already resolved during the course of writing this report. While full deployment and inferencing on the Jetson still requires further profiling, optimization, and long-duration stability validation, the current system demonstrates that a heavily compressed continuous speech-to-speech architecture can be executed with near-reference behaviour on constrained edge hardware. This establishes a strong foundation for future work in real-time embedded speech generation and low-memory neural inference systems.

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